

STA 6707 Final Project
Professor Raid Amin
Group # 2

States:

Michigan, Wisconsin, Minnesota, North Dakota, Montana, Washington

Variables:

Autumn: sixtyfiveandup, Smoking_Rate

Elena: Diabetes, Cancer

Clayton: Mercury_TPY, Lead_TPY

Oscar: Heart_disease, physical_inactivity

Michael Ho: Glyphosates, NATA_Cancer_11

States for PCA and FA:

Autumn: North Dakota

Elena: Washington

Clayton: Michigan

Oscar: Minnesota

Michael Ho: Montana

SAS Group Project for STA6707 spring 2024

Name: Boiko Elena

Introduction:

We considered the data set of Alzheimer's Disease mortality rate for six northern states: Michigan (MI), Minnesota (MN), Montana (MT), North Dakota (ND), Washington (WA), and Wisconsin (WI). The two factors were researched such as Diabetes and Cancer rates. For Principal Component Analysis and Factor Analysis we chose Washington (WA) state and all group variables: sixtyfiveandup, Smoking_Rate, physical_inactivity, Diabetes, Heart_Disease Cancer Glyphosates, NATA_Cancer_11, Mercury_TPY, and Lead_TPY.

1. Importing data:

(a) **Problem:** The data was loaded and filtered by state using the following commands.

(b) **SAS program:**

```
/* Consider the data set on Alzheimer's Disease in the 6 States (MI, MN, MT, ND, WA, WI) */
* 1.1 Reading data;
Proc import datafile = '/home/u63781314/Project/Alz_mortality_data_complete (1).csv'
out=Altz dbms=csv replace;
getnames= yes;
run;
/* Sort data by states */
data Altz;
  set Altz;
  State=scan(County, 2, ',');
  County=scan(County, 1, ',');
  z = _N_;
  if z ge 1198 and z le 1280 then state = 'MI';
  if z ge 1281 and z le 1367 then state = 'MN';
  if z ge 1565 and z le 1620 then state = 'MT';
  if z ge 1957 and z le 2009 then state = 'ND';
  if z ge 2921 and z le 2959 then state = 'WA';
  if z ge 3015 and z le 3086 then state = 'WI';
  if state NOT IN ("MI","MN","MT","ND","WA","WI") then delete;
run;

/* Create a new dataset with only the desired variables */
data Altz;
  set Altz (keep = var1 state sixtyfiveandup Smoking_Rate physical_inactivity Diabetes
  Heart_Disease Cancer Glyphosates NATA_Cancer_11 Mercury_TPY Lead_TPY);
run; /* keep only the desired variables */
```

(c) **SAS results:**

Obs	VAR1	sixtyfiveandup	Smoking_Rate	physical_inactivity	Diabetes	Heart_Disease	Cancer	Glyphosates	NATA_Cancer_11	Mercury_TPY	Lead_TPY	State
1	Alcona County, MI	31.7	26.6	23.68	9.32	153.2	181.9452485	2.03024	18.455	0.000274078	0.009190339	MI
2	Alger County, MI	21.3	26.8	23.2	8.74	115.5	173.088909	0.24425	14.613	0.010142057	0.004728691	MI
3	Allegan County, MI	13.3	23.24	22.82	8.24	98.1	169.940452	48.66074	25.814	0.005582929	0.212753663	MI
4	Alpena County, MI	19.6	25.5	20.78	8.92	148.9	177.2942628	6.17526	21.186	0.100103297	0.064822623	MI
5	Antrim County, MI	22.5	22.88	22.52	8.22	108.1	173.5860126	9.07635	19.263	0.000810993	0.020337146	MI

(d) Comments:

We created a data set of Alzheimer's Disease mortality rate for six northern states and desired variables.

2. Univariate normality assessment

(a) Problem:

The UNIVARIATE procedure was used to test each variable for normality with the Shapiro-Wilk test alongside a QQ-plot, histogram, and boxplot.

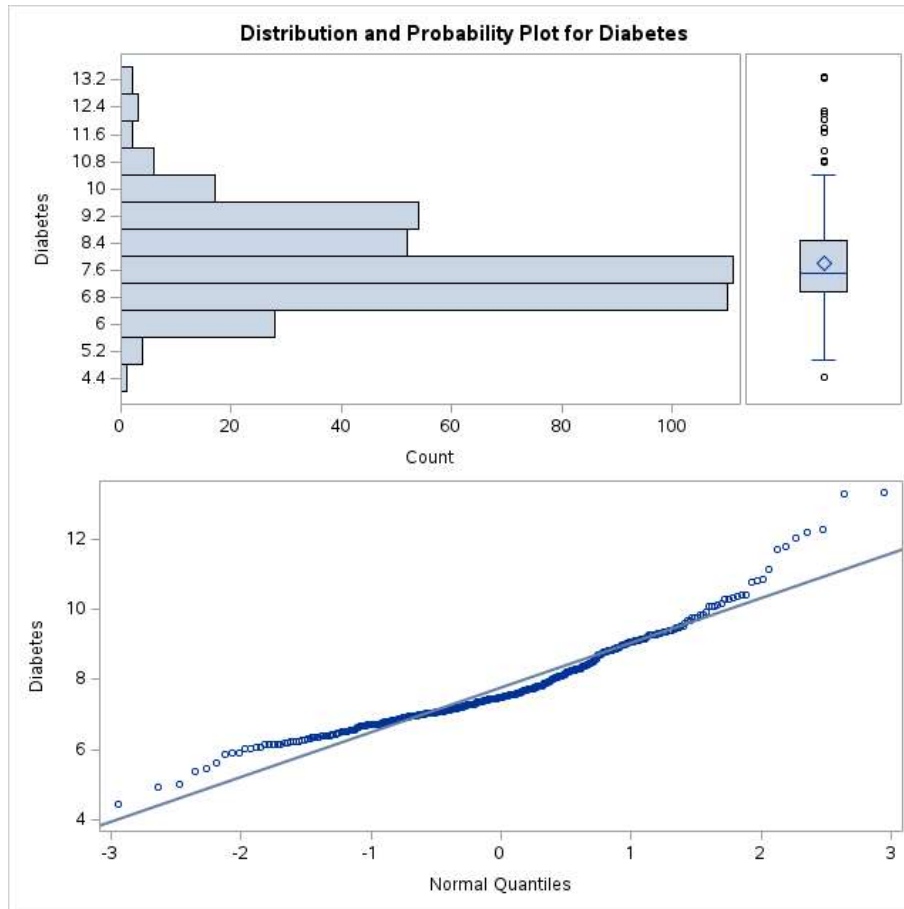
(b) SAS program:

```
/* 2.1 Perform univariate tests for normality */
proc univariate normal plot;
var Diabetes Cancer;
run;
```

2.1 Diabetes

(c) SAS results:

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.934292	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.114124	Pr > D	<0.0100
Cramer-von Mises	W-Sq	1.270415	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	6.904751	Pr > A-Sq	<0.0050



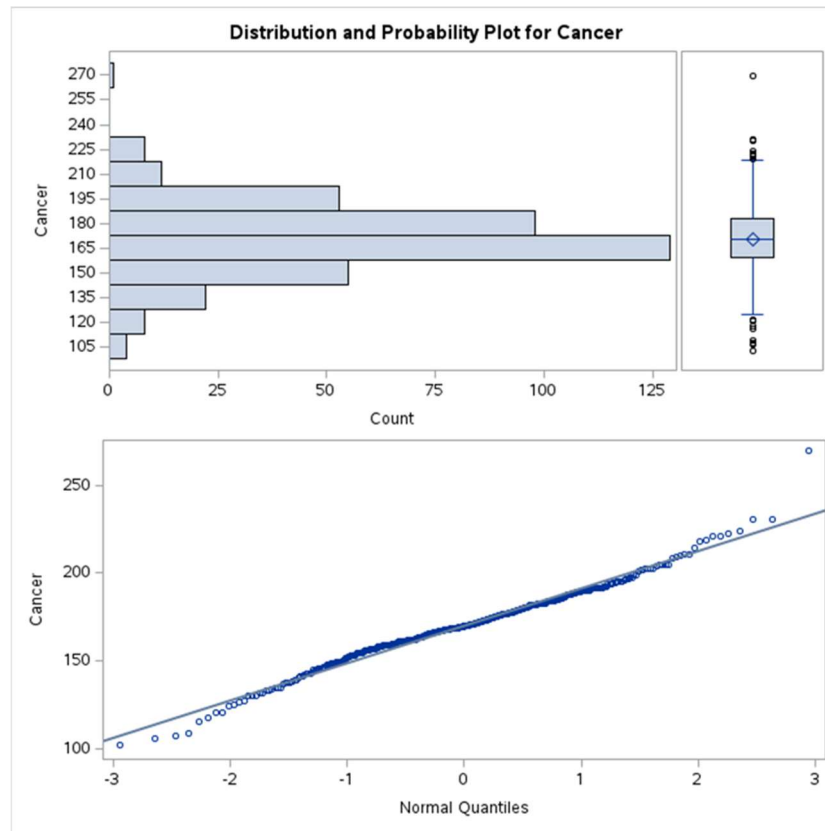
(d) Comments:

Based on the Shapiro-Wilk test (p-value < 0.0001) shown above, we reject the assumption that the random sample of the Diabetes variable follows a normal distribution of data. This conclusion is consistent with the skewness of the QQ plot and histogram. Also, notice a few outliers in the QQ plot and box plot, this one low-value outlier and a few high-value outliers that are mostly situated on the top (the largest values) of the plots.

2.2. Cancer

(c) SAS results:

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.983303	Pr < W	0.0002
Kolmogorov-Smirnov	D	0.053817	Pr > D	<0.0100
Cramer-von Mises	W-Sq	0.231374	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	1.436401	Pr > A-Sq	<0.0050



(d) Comments:

Based on the Shapiro-Wilk test (p-value = 0.0002) shown above, we reject the assumption that the random sample of the Cancer variable follows a normal distribution of data. This conclusion is consistent with the skewness of the QQ plot and histogram. Also, notice a few outliers in the QQ plot and box plot, these few low-value outliers and a few high-value outliers.

2.2 Univariate normality assessment by states

Let's test how variables are distributed by states because it can be beneficial for the next data transformation.

2.2.1 Diabetes

(a) Problem:

The UNIVARIATE procedure was used to test Diabetes data for normality by states with the Shapiro-Wilk test alongside a QQ-plot, histogram, and box-plot.

(b) SAS program:

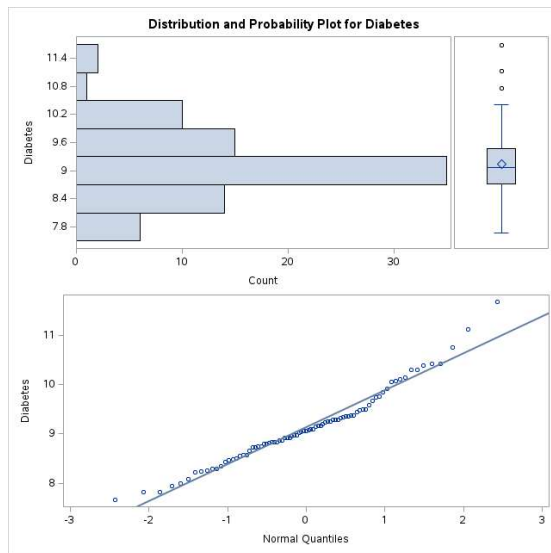
```
/* 2.2.1 Perform univariate tests for normality by states for Diabetes*/
proc sort; by state;
proc univariate normal plot;
var Diabetes; by state; run;
```

(c) SAS results:

MI

Tests for Normality

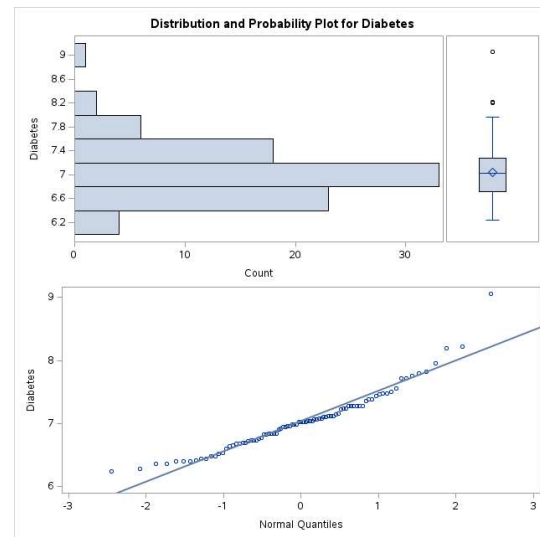
Test	Statistic	p Value		
Shapiro-Wilk	W	0.966161	Pr < W	0.0276
Kolmogorov-Smirnov	D	0.108961	Pr > D	0.0162
Cramer-von Mises	W-Sq	0.144692	Pr > W-Sq	0.0278
Anderson-Darling	A-Sq	0.798009	Pr > A-Sq	0.0390



MN

Tests for Normality

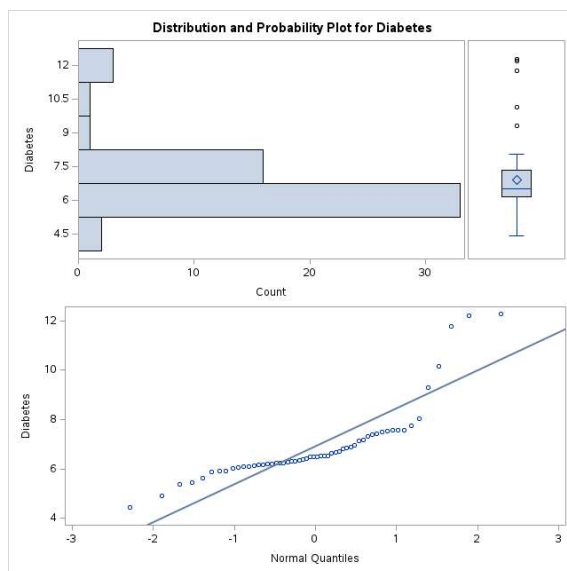
Test	Statistic	p Value		
Shapiro-Wilk	W	0.9352	Pr < W	0.0003
Kolmogorov-Smirnov	D	0.10163	Pr > D	0.0248
Cramer-von Mises	W-Sq	0.144949	Pr > W-Sq	0.0275
Anderson-Darling	A-Sq	0.962299	Pr > A-Sq	0.0159



MT

Tests for Normality

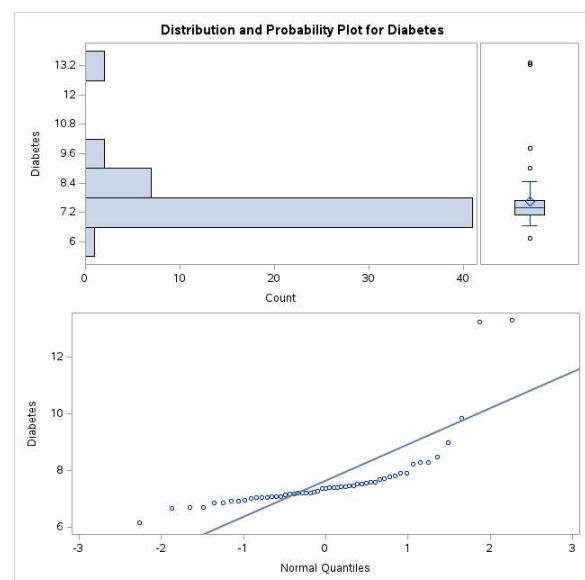
Test	Statistic	p Value		
Shapiro-Wilk	W	0.752324	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.209903	Pr > D	<0.0100
Cramer-von Mises	W-Sq	0.801112	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	4.616787	Pr > A-Sq	<0.0050



ND

Tests for Normality

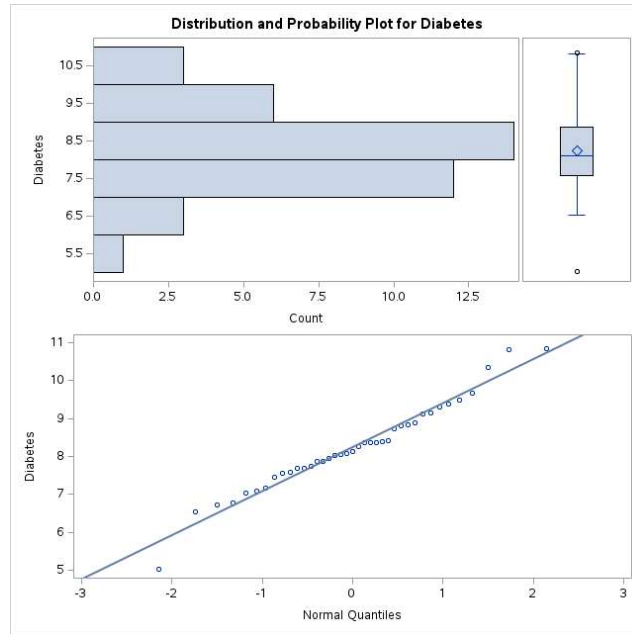
Test	Statistic	p Value		
Shapiro-Wilk	W	0.586418	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.270622	Pr > D	<0.0100
Cramer-von Mises	W-Sq	1.21393	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	6.693037	Pr > A-Sq	<0.0050



WA

Tests for Normality

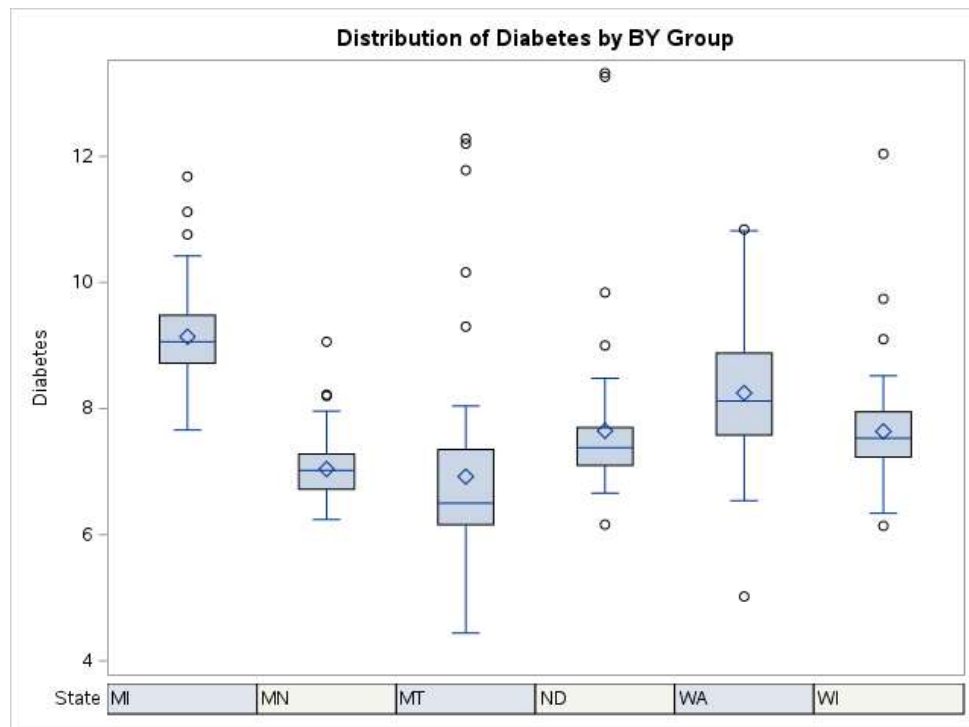
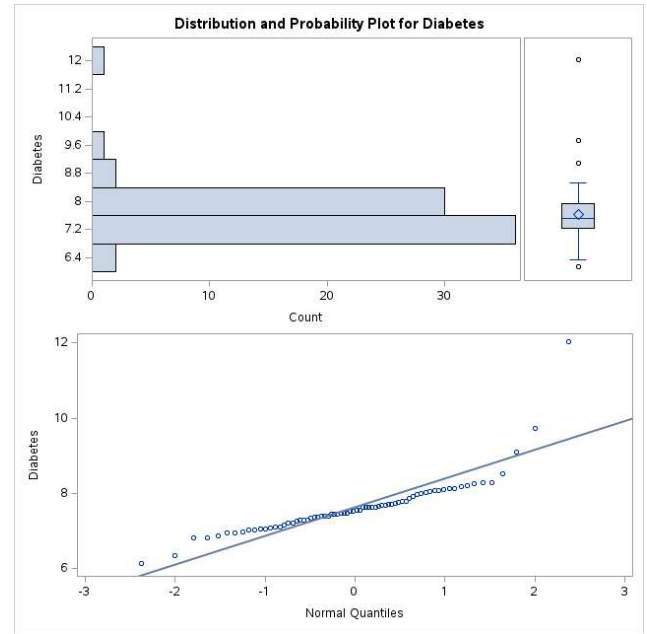
Test	Statistic	p Value
Shapiro-Wilk	W	0.974637 Pr < W 0.5142
Kolmogorov-Smirnov	D	0.107327 Pr > D >0.1500
Cramer-von Mises	W-Sq	0.049852 Pr > W-Sq >0.2500
Anderson-Darling	A-Sq	0.343189 Pr > A-Sq >0.2500



WI

Tests for Normality

Test	Statistic	p Value
Shapiro-Wilk	W	0.770542 Pr < W <0.0001
Kolmogorov-Smirnov	D	0.145177 Pr > D <0.0100
Cramer-von Mises	W-Sq	0.4738 Pr > W-Sq <0.0050
Anderson-Darling	A-Sq	3.043365 Pr > A-Sq <0.0050



(d) Comments:

The only one of the six states where Diabetes data is normally distributed is Washington state(WA). Based on the result of the Shapiro-Wilk test (p-value=0.5142) shown above we accept the normality assumption. The next 5 states show that the random sample Diabetes doesn't follow a normal data distribution: the Shapiro-Wilk test is deficient (<0.05) for MI (p-value=**0.0276**), MN (p-value=**0.0003**), MT, ND, and WI (each p-value <0.0001). This conclusion is corroborated by the skewness of the Q-Q plot and histogram shown below. Also, notice the few outliers in each state indicated in the Q-Q plot and box plot. The most distinct are the high-value outliers and they probably create the main problem for analysis.

The box plot of the distribution of Diabetes by states (group) supports this conclusion by the wide difference in means values and many outliers' presence.

2.2.2 Cancer

(a) Problem:

The UNIVARIATE procedure was used to test Cancer data for normality by states with the Shapiro-Wilk test alongside a QQ-plot, histogram, and box-plot.

(b) SAS program:

```
/* 2.2.2 Perform univariate tests for normality by states for Cancer*/
```

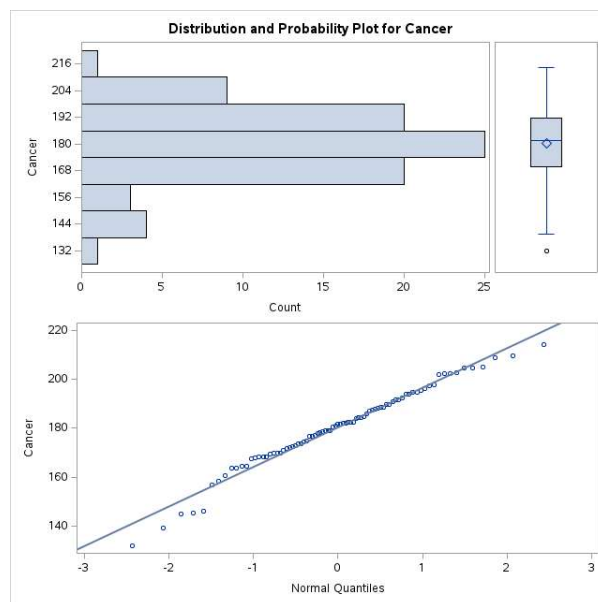
```
proc sort; by state;
proc univariate normal plot;
var Cancer; by state; run;
```

(c) SAS results:

MI

Tests for Normality

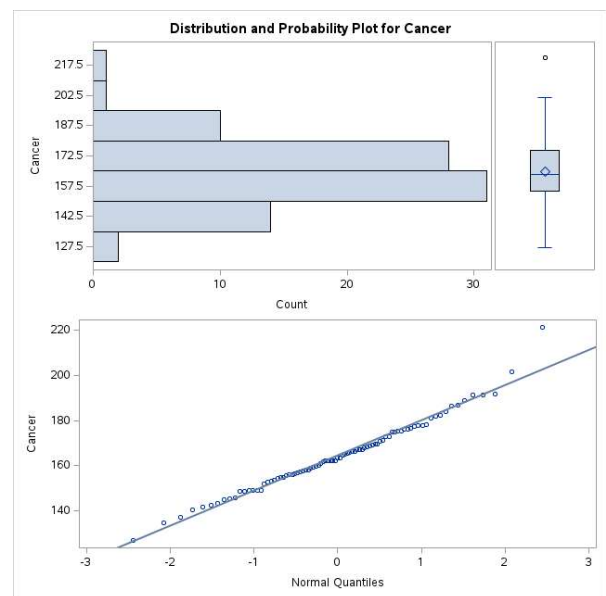
Test	Statistic	p Value		
Shapiro-Wilk	W	0.977941	Pr < W	0.1642
Kolmogorov-Smirnov	D	0.070053	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.043729	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.403743	Pr > A-Sq	>0.2500



MN

Tests for Normality

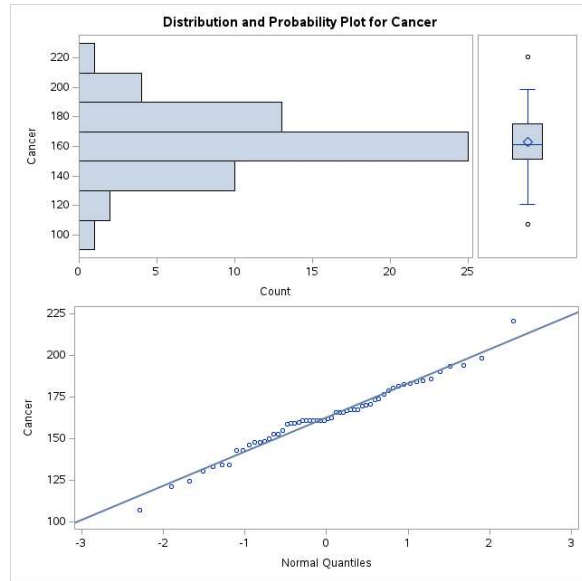
Test	Statistic	p Value		
Shapiro-Wilk	W	0.980949	Pr < W	0.2289
Kolmogorov-Smirnov	D	0.057881	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.045242	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.300584	Pr > A-Sq	>0.2500



MT

Tests for Normality

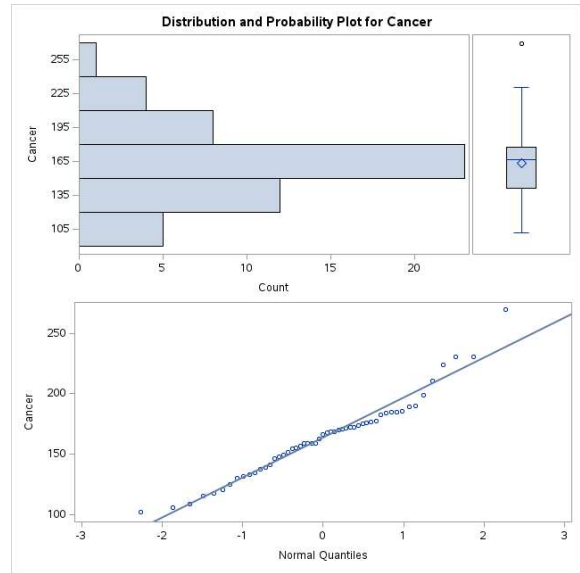
Test	Statistic	p Value			
Shapiro-Wilk	W	0.982802	Pr < W	0.6037	
Kolmogorov-Smirnov	D	0.12117	Pr > D	0.0399	
Cramer-von Mises	W-Sq	0.088462	Pr > W-Sq	0.1603	
Anderson-Darling	A-Sq	0.457841	Pr > A-Sq	>0.2500	



ND

Tests for Normality

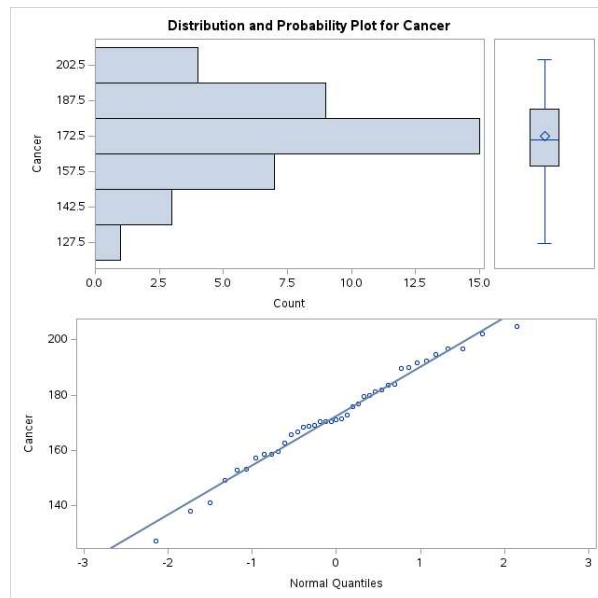
Test	Statistic	p Value			
Shapiro-Wilk	W	0.962164	Pr < W	0.0915	
Kolmogorov-Smirnov	D	0.105788	Pr > D	0.1427	
Cramer-von Mises	W-Sq	0.080933	Pr > W-Sq	0.2047	
Anderson-Darling	A-Sq	0.538465	Pr > A-Sq	0.1659	



WA

Tests for Normality

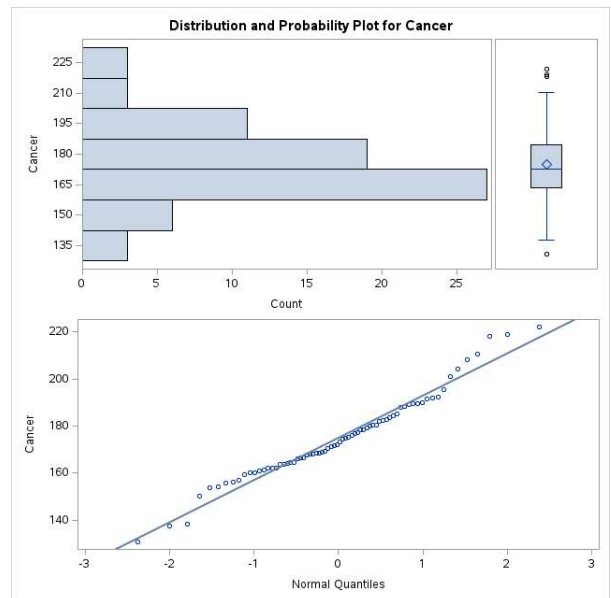
Test	Statistic	p Value			
Shapiro-Wilk	W	0.981842	Pr < W	0.7697	
Kolmogorov-Smirnov	D	0.079451	Pr > D	>0.1500	
Cramer-von Mises	W-Sq	0.031641	Pr > W-Sq	>0.2500	
Anderson-Darling	A-Sq	0.212826	Pr > A-Sq	>0.2500	

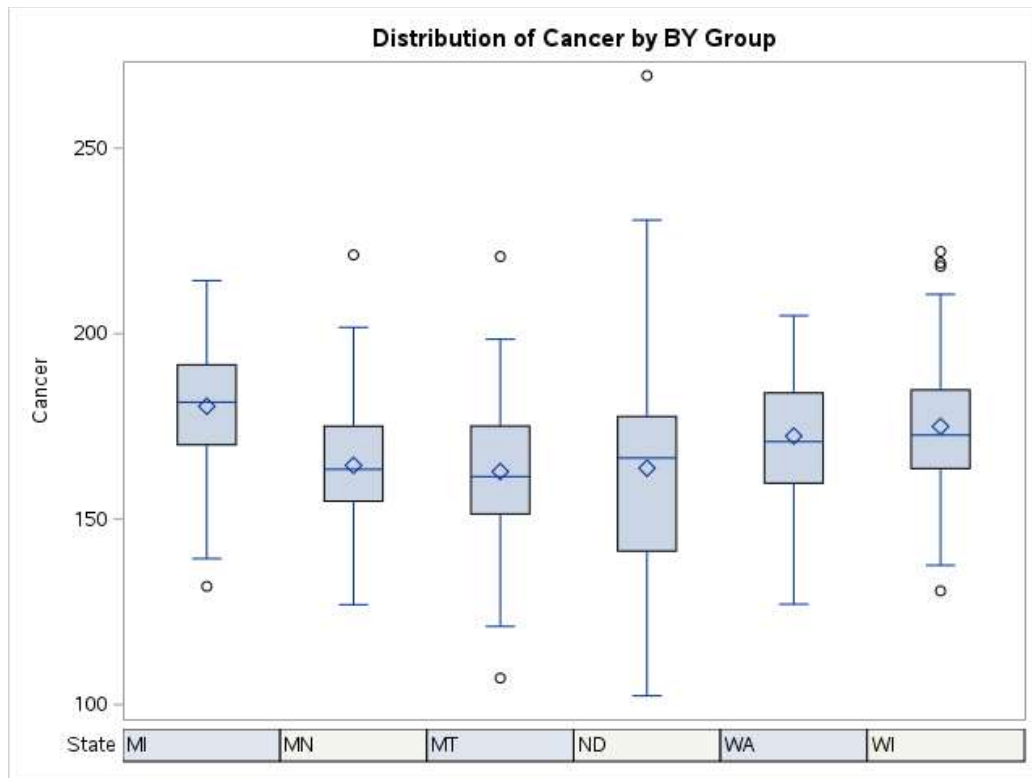


WI

Tests for Normality

Test	Statistic	p Value			
Shapiro-Wilk	W	0.971156	Pr < W	0.0968	
Kolmogorov-Smirnov	D	0.068319	Pr > D	>0.1500	
Cramer-von Mises	W-Sq	0.10048	Pr > W-Sq	0.1109	
Anderson-Darling	A-Sq	0.699626	Pr > A-Sq	0.0685	





(d) Comments:

All six states show that the random sample Diabetes follows a normal data distribution: the Shapiro-Wilk test is > 0.05 for MI (p-value=**0.1642**), MN (p-value=**0.2289**), MT (p-value=**0.6037**), ND (p-value=**0.0915**), WA (p-value=**0.7697**), and WI (p-value=**0.0968**). This conclusion is corroborated by the very light skewness of the Q-Q plot and histogram shown above. Also, notice the rare outliers in some states indicated in the Q-Q plot and box plot.

This is interesting because Cancer data is normal in each state, but in all six states together Shapiro-Wilks shows 0.002, and we reject the normality assumption. This phenomenon is known as Simpson's paradox. Simpson's paradox occurs when a trend appears in several different groups of data but disappears or reverses when the groups are combined. This can happen due to confounding variables or differences in the compositions of the groups.

The box plot of the distribution of Cancer by states (group) shows the small difference in mean values and rare outliers' presence, which supports our conclusion about approximate normality.

2.3 Outliers assessment

Let's test how Diabetes outliers are distributed by states and counties because it can be beneficial for the next data transformation. Cancer outliers don't look problematic in this stage of analysis.

(a) **Problem:** Using **proc.sort** identify the counties with extremely large or small Diabetes values.

(b) **SAS program:**

```

/* Sort the data by the smallest Diabetes variable */
proc sort data=Altz;
  by Diabetes;
run;
/* Print the top and bottom few observations to identify counties with extreme small values */
proc print data=Altz(obs=10) ; /* Print the first 10 observations */
  title 'Counties with Smallest Diabetes Values';
run;
/* Sort the data by the largest Diabetes variable in descending order */
proc sort data=Altz;
  by descending Diabetes;
run;
/* Print the top and bottom few observations to identify counties with extreme large values */
proc print data=Altz(obs=10); /* Print the first 10 observations */
  title 'Counties with Largest Diabetes Values';
run;

```

(c) SAS results:

Counties with Largest Diabetes Values

Obs	VAR1	sixtyfiveandup	Smoking_Rate	physical_inactivity	Diabetes	Heart_Disease	Cancer	Glyphosates	NATA_Cancer_11	Mercury_TPY	Lead_TPY	State
1	Sioux County, ND	7.1	36.04	31.06	13.32	167.7	269.5572498	24.07352	15.434	0.0000391	0.000405825	ND
2	Rolette County, ND	10	31.84	34.04	13.26	116.1	223.7327475	50.33502	14.594	0.000249114	0.01531556	ND
3	Roosevelt County, MT	11.3	32.62	31.12	12.28	188.5	170.1863854	97.90982	15.784	0.00210021	0.04702075	MT
4	Big Horn County, MT	10.5	31.28	27.76	12.2	103.6	154.8709818	31.93188	18.205	0.005138637	0.032517402	MT
5	Menominee County, WI	13.3	34.68	30.26	12.04	125.2	219.0781438	0.00036	21.174	0.000130539	0.0000363	WI
6	Glacier County, MT	10.6	31.2	29.92	11.78	123.7	184.0348068	92.65241	14.671	0.002254688	0.020276846	MT
7	Wayne County, MI	12.7	26.86	26.62	11.68	178.4	201.8189055	2.50839	37.599	0.2705518	1.338751748	MI
8	Genesee County, MI	13.9	26.42	28.22	11.12	150	197.2648708	32.82371	32.122	0.020073594	0.249749876	MI
9	Grays Harbor County,	16.5	27.62	24.24	10.84	130.1	202.2665233	9.09161	17.891	0.008162312	0.281062473	WA
10	Garfield County, WA	25.3	19.08	25.22	10.82	99.51288837	137.8563636	21.27323	16.968	0.0000469	0.00181948	WA

Counties with Smallest Diabetes Values

Obs	VAR1	sixtyfiveandup	Smoking_Rate	physical_inactivity	Diabetes	Heart_Disease	Cancer	Glyphosates	NATA_Cancer_11	Mercury_TPY	Lead_TPY	State
1	Gallatin County, MT	9.5	14.7	15.96	4.44	67.4	133.3246456	15.50192	21.767	0.009920184	0.624861779	MT
2	Missoula County, MT	11.4	19.14	16.92	4.92	78.5	165.997775	0.59303	29.939	0.004711183	0.084137654	MT
3	San Juan County, WA	23.5	16.04	12.9	5.02	49.3	127.0802205	0.19954	21.087	0.000230079	0.426107395	WA
4	Ravalli County, MT	19.5	21.2	18.12	5.38	75.9	150.1987626	1.82125	18.087	0.000641694	0.105111647	MT
5	Park County, MT	17	19.52	19.14	5.46	68.5	147.6363448	3.40999	15.522	0.000490266	0.048178196	MT
6	Madison County, MT	21.2	20.64	24.8	5.64	76.9	146.4043763	4.10598	12.881	0.001440325	0.038763572	MT
7	Flathead County, MT	14.5	20.26	18.66	5.88	97.1	152.7198026	4.59118	23.541	0.003818514	0.21505956	MT
8	Garfield County, MT	26	23.5	26.1	5.9	89.13634027	160.9167967	51.99561	15.615	0.0000821	0.008248203	MT
9	Granite County, MT	25.3	22.02	23.42	5.9	117.2	107.1495656	0.61561	15.699	0.000302445	0.007574979	MT
10	Powder River County,	22.1	21.88	28.38	6.02	89.13634027	142.965538	11.81945	17.724	0.0000888	0.014654039	MT

(d) Comments:

We sorted outliers by states and counties and chose the ten largest and ten smallest values hoping to identify the preferred state or county. Unfortunately, when the low-value outliers came mostly from one state MT, the high-value outliers came from four different states, MI, MT, ND, and WA. So, we did not find the main state or county for transformation.

2.4 Bivariate test for normality

A scatter plot of the pairs of variables as well as a bivariate Gamma plot (chi-squared Q-Q plot) can be used in checking bivariate normality (Johnson & Wichern, 2007). However, correlation doesn't necessarily indicate bivariate normality. It only measures the linear relationship between pairs of variables.

(a) Problem:

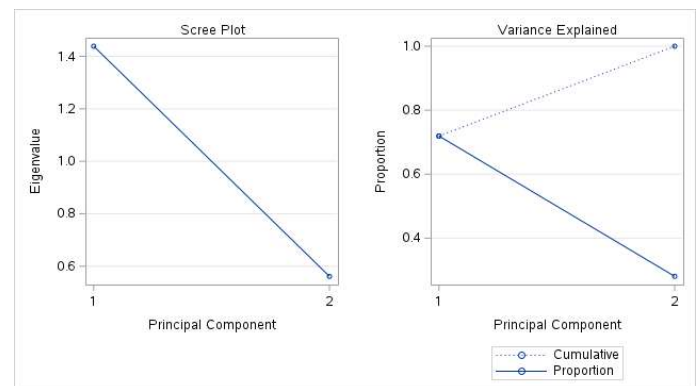
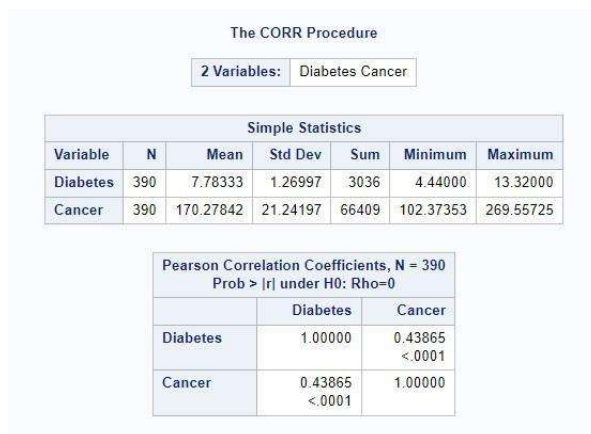
Use bivariate tests for normality for Diabetes and Cancer to inspect the relationships between variables and their distributions.

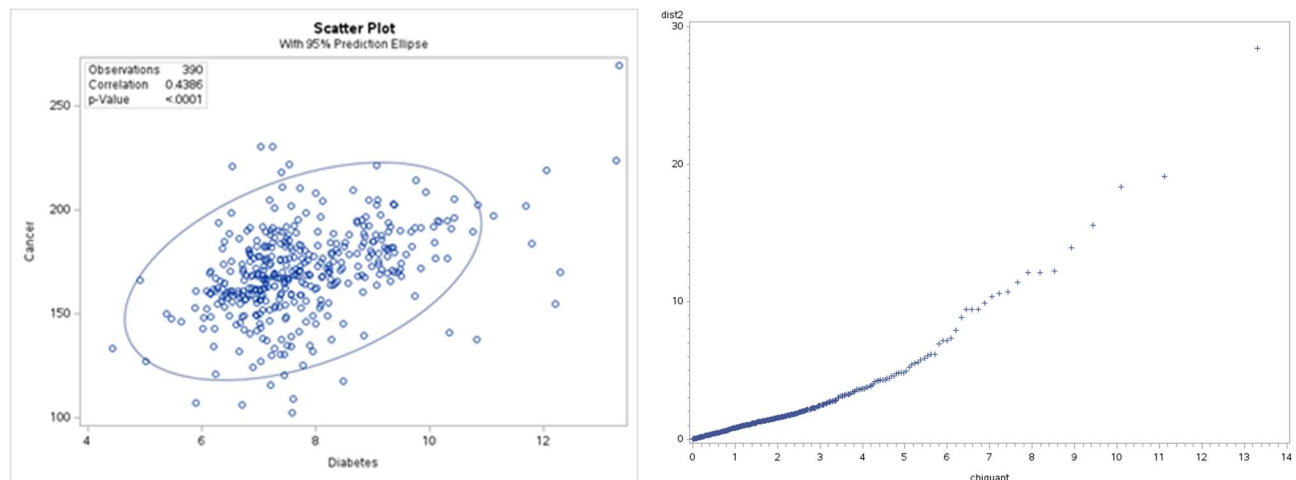
(b) SAS program:

```
/* 2.4 Bivariate tests for normality */
/* Generate scatterplots for pairs of variables */;
proc corr data=Altz plots(only)=scatter;
    var Diabetes Cancer;
run;

* Chi-Square plot for Diabetes and Cancer;
proc princomp std out=pcresult;var Diabetes Cancer;
run;
data mahal;set pcresult;dist2=uss(of prin1-prin2);
run;
proc sort;by dist2;run;
data plotdata;set mahal;
prb=(_n_-.5)/390;chiquant=cinv(prb,2);
run;
proc gplot;plot dist2*chiquant;
run;
```

(c) SAS results:





(d) Comments:

Most of the points are found inside the ellipse from the scatter plot, but some points are out from the ellipse. Likewise, for the Gamma plot, most of the points fall a little lower to the straight line with deviations of outliers at the last few points in the top right being apparent outliers. Thus, for both plots, some points are from the ellipse (for scatter plot) and the straight line (Gamma plot). Based on visual inspection we cannot conclude bivariate normality.

The Pearson correlation coefficient between the two variables is 0.43865, which means that the moderate strength of the linear relationship between Diabetes and Cancer is present.

3. Finding transformation for normality.

3.1 Box-Cox transformation.

Since the data for Diabetes and Cancer is all greater than zero, it is suitable for the Box-Cox family of power transformation. We will search for an optimal λ (lambda) using the TRANSREG procedure to perform a Box-Cox analysis for each variable, transform the data using the optimal λ , and use Shapiro-Wilk test to compare the new Wilk test p -value with the corresponding raw data p -value. The

transformation is applied using the Box-Cox transformation, which uses the form $\frac{(y^\lambda - 1)/\lambda}{\log(y)}$ $\lambda \neq 0$
 $\lambda = 0$,
 where Y is the raw variable.

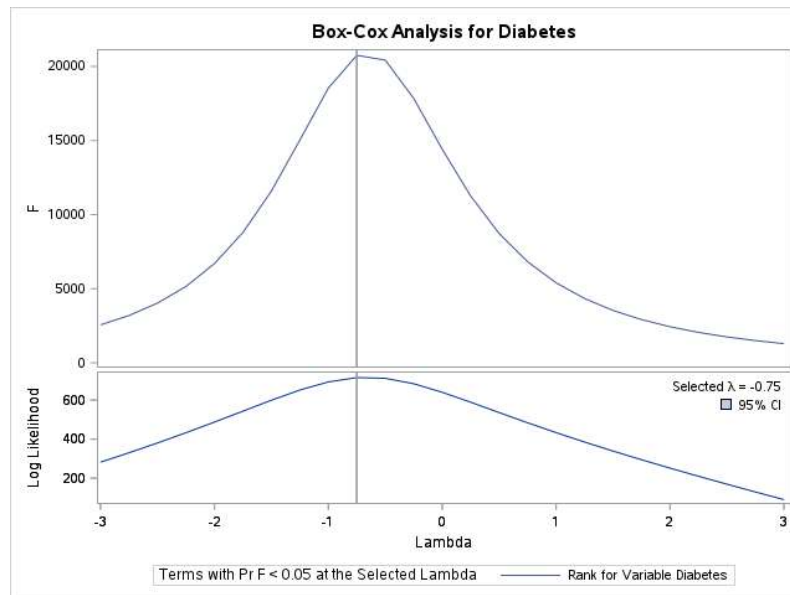
3.1.1 Box-Cox transformation. Diabetes

(a) 1. Problem: Using TRANSREG gives us an optimal lambda for Diabetes variable based on the normal quantiles.

(b) SAS program:

```
/* 3.1.1 Perform box-cox transformation for Diabetes*/
proc rank normal=blom out=normals;
var Diabetes; ranks Dq;
proc transreg;
model boxcox(Diabetes) = identity(Dq);
run;
```

(c) SAS results:



(d) Comments: The optimal lambda for Diabetes is chosen as **-0.75** resulting in a transformation

(a) 2. Problem:

Apply Box-Cox transformation using the optimal $\lambda = -0.75$ and Shapiro-Wilk test to compare the new Wilk test p -value with the corresponding raw data p -value.

(b) SAS program:

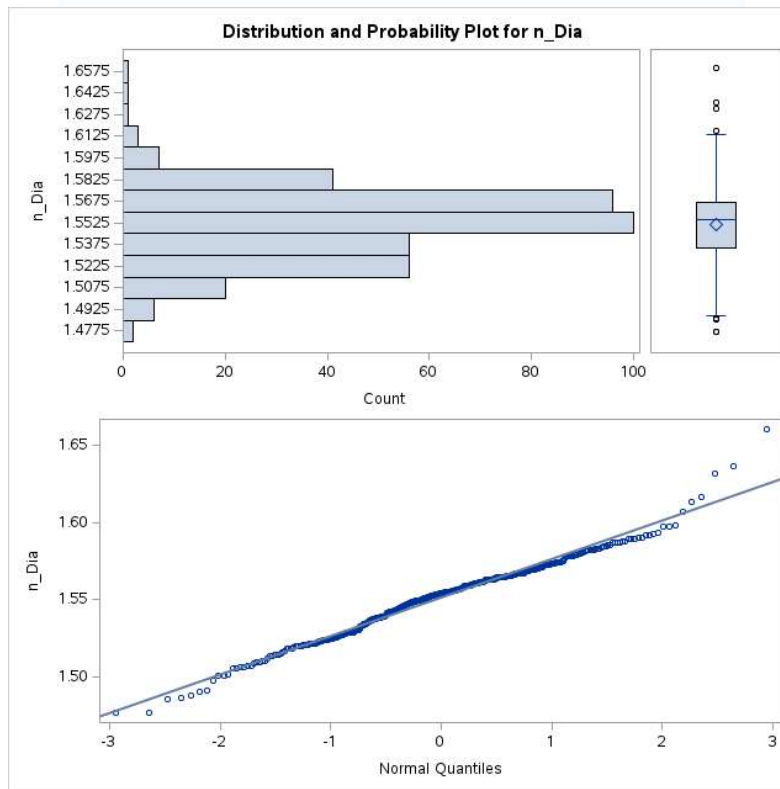
```
/* Apply new transformation */
data Altz;
set Altz;
n_Dia = (Diabetes**(-0.75)-1/(-0.75));

/*print the data to the screen to see if they are okay;
proc print data= Altz;
run;

/* Check for normality after transformation */
proc univariate normal plot;
var n_Dia; run;
```

(c) SAS results:

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.983303	Pr < W	0.0002
Kolmogorov-Smirnov	D	0.053817	Pr > D	<0.0100
Cramer-von Mises	W-Sq	0.231374	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	1.436401	Pr > A-Sq	<0.0050



(d) Comments:

We see that Wilk test p-values (0.0002) increased negligibly, and still imply that Diabetes data does not follow the normal distribution across the six northern states. Also, we see a smaller number of outliers in the QQ plot and box plot, but they are still present.

3.1.2 Box-Cox transformation. Cancer.

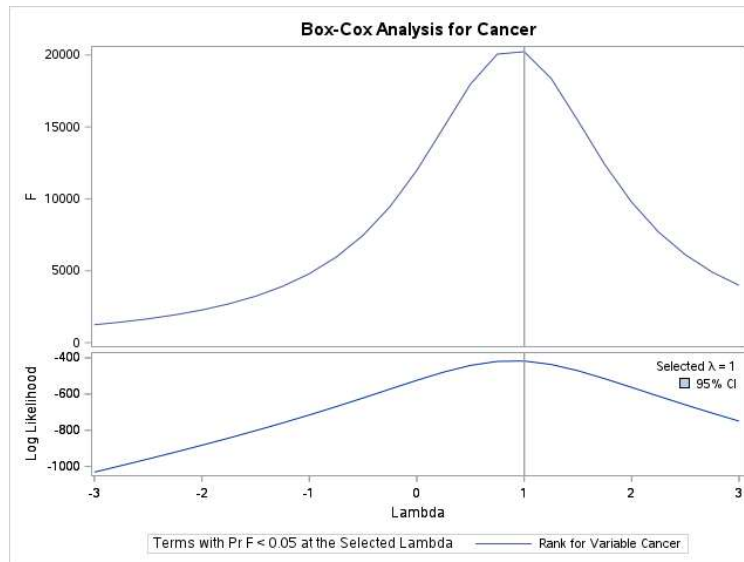
(a) 1. Problem: Using TRANSREG gives us an optimal lambda for Cancer variable based on the normal quantiles.

(b) SAS program:

```
/* 3.1.2 Perform box-cox transformation for Cancer*/
proc rank normal=blom out=normals;
var Cancer; ranks Cq;
proc transreg;
```

```
model boxcox(Cancer) = identity(Cq);
run;
```

(c) SAS results:



(d) Comments:

The optimal lambda for Cancer is chosen as 1. Note that using this lambda with the Box-Cox transformation formula will not have any effect on the distribution of the data. This is interesting considering that Cancer data has a low p-value=0.0002 in the Wilk test showing that this is not a normally distributed variable. It may be the case that Cancer does not significantly benefit from any power transformation, and applying transformations can complicate analysis and the interpretation of results.

3.2 Quantile transformation.

(a) Problem:

Perform a quantile transformation on our variable Diabetes and Cancer to achieve approximate normality using the PROC RANK procedure. Create a transformed data set before conducting further analysis.

(b) SAS program:

```
/* 3.2 Quantile transformation for variables */
proc rank data=Altz normal=blom out=normals;
  var Diabetes;
  ranks new_Diabetes;
  var Cancer;
  ranks new_Cancer; run;
/* Perform univariate tests for normality for transformed Diabetes */
proc univariate data=normals normal plot;
  var new_Diabetes;
run;
```

```
/* Perform univariate tests for normality for transformed Cancer */
```

```
proc univariate data=normals normal plot;
```

```
var new_Cancer;
```

```
run;
```

```
/* Create new dataset with transformed variable */
```

```
data Altz_transformed;
```

```
merge Altz(in=a) normals(in=b);
```

```
run;
```

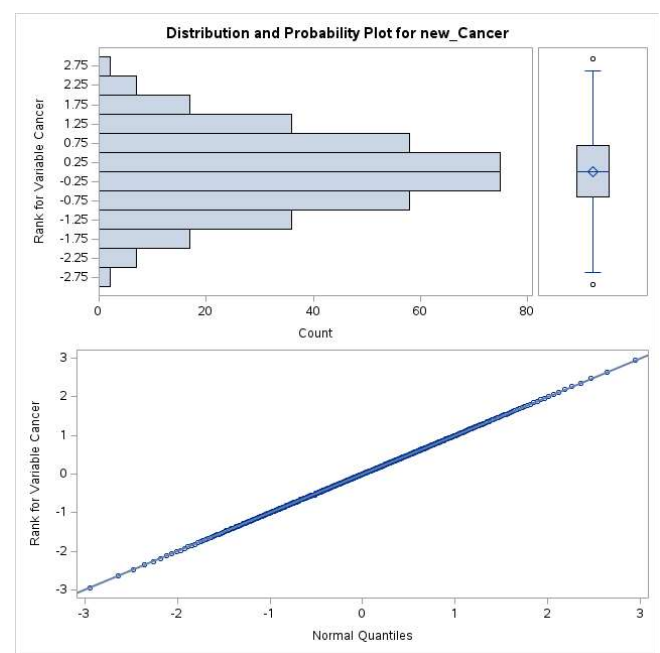
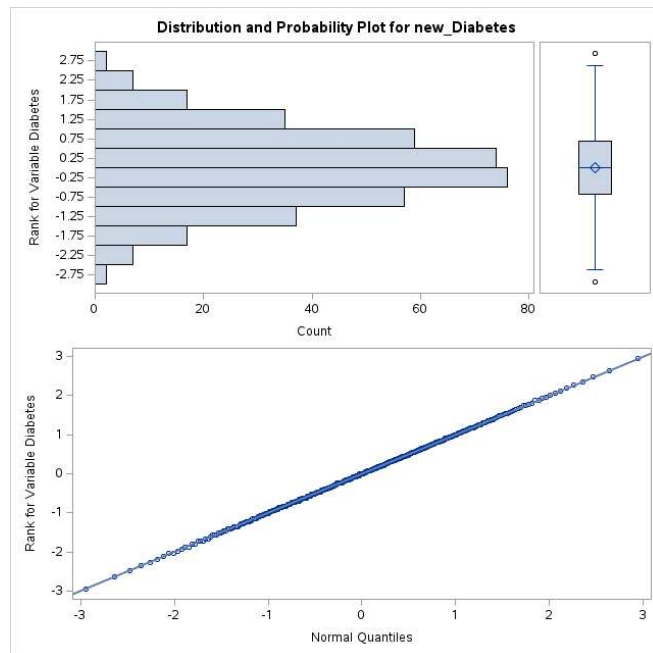
(c) SAS results:

Transformed new_Diabetes data

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.999677	Pr < W	1.0000
Kolmogorov-Smirnov	D	0.011824	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.002916	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.018684	Pr > A-Sq	>0.2500

Transformed new_Cancer data

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.999754	Pr < W	1.0000
Kolmogorov-Smirnov	D	0.009527	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.000515	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.005327	Pr > A-Sq	>0.2500



Transformed Data														
Obs	VAR1	sixtyfiveandup	Smoking_Rate	physical_inactivity	Diabetes	Heart_Disease	Cancer	Glyphosates	NATA_Cancer_11	Mercury_TPY	Lead_TPY	State	new_Diabetes	new_Cancer
1	Alcona County, MI	31.7	26.6	23.68	9.32	153.2	181.9452485	2.03024	18.455	0.000274078	0.009190339	MI	1.23066	0.59541
2	Alger County, MI	21.3	26.8	23.2	8.74	115.5	173.088909	0.24425	14.613	0.010142057	0.004728691	MI	0.77853	0.17756
3	Allegan County, MI	13.3	23.24	22.82	8.24	98.1	169.940452	48.66074	25.814	0.005582929	0.212753663	MI	0.53883	0.00963
4	Alpena County, MI	19.6	25.5	20.78	8.92	148.9	177.2942628	6.17526	21.186	0.100103297	0.064822623	MI	0.90200	0.37815
5	Antrim County, MI	22.5	22.88	22.52	8.22	108.1	173.5860126	9.07635	19.263	0.000810993	0.020337146	MI	0.52036	0.19718
6	Arenac County, MI	20.6	28	22.44	9.08	200.8	204.8328321	17.18994	20.626	0.000328767	0.000236343	MI	1.01376	1.66732
7	Baraga County, MI	17.3	28.6	21.84	10.14	167.9	194.8706807	0.08301	16.642	0.002123738	0.016336399	MI	1.66732	1.31741
8	Barry County, MI	14.7	23.44	23.1	8.34	104.8	164.6296331	29.87018	25.29	0.001231001	0.107006348	MI	0.61858	-0.33028
9	Bay County, MI	16.5	25.04	23.6	9.16	164.4	187.6308219	54.91432	29.548	0.075012971	0.189653507	MI	1.09207	0.88284
10	Benzie County, MI	21.1	21.32	20.68	8.82	106.3	160.5154355	2.49774	18.151	0.000309739	0.017193364	MI	0.81828	-0.56501
11	Berrien County, MI	16.2	24.54	26.7	10.08	114.5	184.4694837	35.23695	28.395	0.009400075	0.175517792	MI	1.61782	0.77419
12	Branch County, MI	15	27.02	26.26	9.5	143.1	174.3036993	65.9104	26.569	0.001705418	0.057647764	MI	1.38969	0.22344
13	Calhoun County, MI	15	26.34	25.34	10.38	125.8	190.8613399	62.94711	30.792	0.01227622	0.318803953	MI	1.81303	1.08043
14	Cass County, MI	16.5	25.5	25.12	8.92	140.8	192.5102548	42.58541	26	0.0416947	0.013863504	MI	0.90200	1.21708
15	Charlevoix County, MI	18.7	23.06	20.1	8.46	99	172.0747044	3.25163	19.477	0.133661633	0.194422632	MI	0.67399	0.13206
16	Cheboygan County, MI	21.8	26.62	25.1	8.72	140.9	193.9140687	0.78836	18.175	0.000855783	0.030765037	MI	0.75698	1.25853
17	Chippewa County, MI	14.6	25.58	24.18	10.3	144.5	176.8932018	0.35547	20.201	0.001089646	0.062846251	MI	1.73576	0.35753
18	Clare County, MI	20.1	28.62	27.82	9.06	178.7	202.4026422	3.07498	20.992	0.006888587	0.048278872	MI	0.99255	1.57200

(d) Comments:

The Shapiro-Wilk test is commonly used to assess normality in data. A small value of W and a p-value less than 0.05 typically indicate a departure from normality, while a p-value greater than 0.05 suggests normality. In our case, the Shapiro-Wilk test yielded p-values of 1.000 for both variables, indicating no departure from normality.

This conclusion is further supported by examining histograms and QQ plots, which show a bell-shaped curve and a linear distribution of points respectively, indicative of normality. Additionally, there are only two extreme values for each variable: one high and one low.

Through quantile transformation, we have successfully achieved approximate normality for both variables. Consequently, we have created a transformed dataset with both raw and transformed variables, ready for further analysis.

Given that our new variables are now normally distributed and each univariate Wilk's lambda test yields a value of 1.0, our data meet the assumptions of multivariate normality for MANOVA. Therefore, there may not be a need to test for bivariate normality in this scenario separately.

4. MANOVA testing

(a) Problem:

Using **proc.glm**, assess whether there are significant differences in the mean values of the transformed variables of Diabetes and Cancer across different states. If there is a significant overall effect of state on the variables, perform the individual pairwise comparisons.

(b) SAS program:

```
/* MANOVA testing for transformed variables */
proc glm;
class state;
model new_Diabetes new_Cancer = state/ss1 ss2;
means state / tukey lines;
means state / tukey;
MANOVA h=state/printe printh ;
Run;
```

4.1 MANOVA

(c) SAS results:

Multivariate

The GLM Procedure					
Multivariate Analysis of Variance					
H = Type II SSCP Matrix for State					
	new_Diabetes	new_Cancer			
new_Diabetes	179.91571608	84.131139427			
new_Cancer	84.131139427	47.477401379			

Characteristic Roots and Vectors of: E Inverse * H, where H = Type II SSCP Matrix for State E = Error SSCP Matrix			
Characteristic Root	Percent	Characteristic Vector V'EV=1	
		new_Diabetes	new_Cancer
0.87399474	97.03	0.06845599	0.00265103
0.02675153	2.97	-0.02691063	0.05733787

MANOVA Test Criteria and F Approximations for the Hypothesis of No Overall State Effect H = Type II SSCP Matrix for State E = Error SSCP Matrix					
S=2 M=1 N=190.5					
Statistic	Value	F Value	Num DF	Den DF	Pr > F
Wilks' Lambda	0.51971623	29.65	10	766	<.0001
Pillai's Trace	0.49243510	25.09	10	768	<.0001
Hotelling-Lawley Trace	0.90074626	34.44	10	571.76	<.0001
Roy's Greatest Root	0.87399474	67.12	5	384	<.0001
NOTE: F Statistic for Roy's Greatest Root is an upper bound.					
NOTE: F Statistic for Wilks' Lambda is exact.					

Univariate

The GLM Procedure					
Dependent Variable: new_Diabetes Rank for Variable Diabetes					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	179.9157161	35.9831432	66.98	<.0001
Error	384	206.2810193	0.5371902		
Corrected Total	389	386.1967354			

R-Square	Coeff Var	Root MSE	new_Diabetes Mean
0.465865	-1.2721E17	0.732933	-0.000000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
State	5	179.9157161	35.9831432	66.98	<.0001

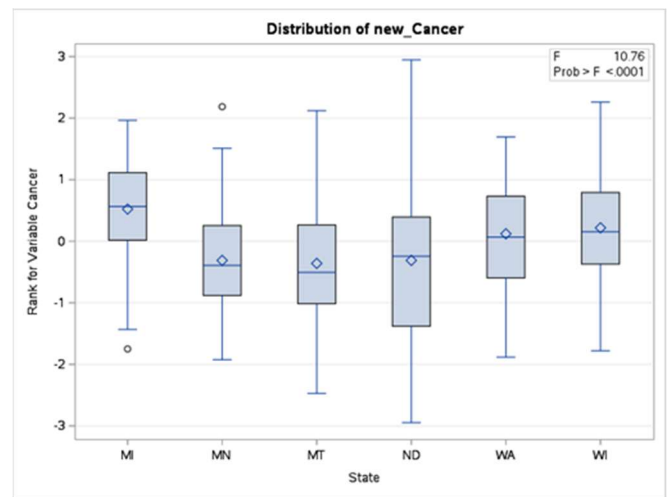
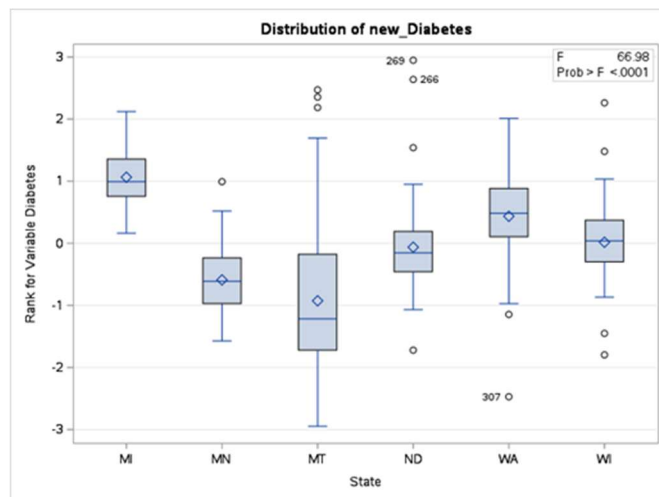
Source	DF	Type III SS	Mean Square	F Value	Pr > F
State	5	179.9157161	35.9831432	66.98	<.0001

The GLM Procedure					
Dependent Variable: new_Cancer Rank for Variable Cancer					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	47.4774014	9.4954803	10.76	<.0001
Error	384	338.7493149	0.8821597		
Corrected Total	389	386.2267163			

R-Square	Coeff Var	Root MSE	new_Cancer Mean
0.122926	-8.4599E17	0.939234	-0.000000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
State	5	47.47740138	9.49548028	10.76	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
State	5	47.47740138	9.49548028	10.76	<.0001



(d) Comments:

The null hypothesis for MANOVA is that there are no differences in the mean vectors of the variables 'Cancer' and 'Diabetes' across the different states.

- **H₀:** There is no significant difference in the combined mean vectors of new_Diabetes and new_Cancer across different states.
- **H_A:** At least one state significantly influences the combined mean vectors of new_Diabetes and new_Cancer.

All p-values are less than **0.0001**, which is below the significance level of **0.05**. Therefore, we reject the null hypothesis and conclude that there are differences in the mean vectors of the transformed variables of Cancer and Diabetes across the different northern states. In other words, the combined mean vector of Cancer and Diabetes is not the same across all northern states.

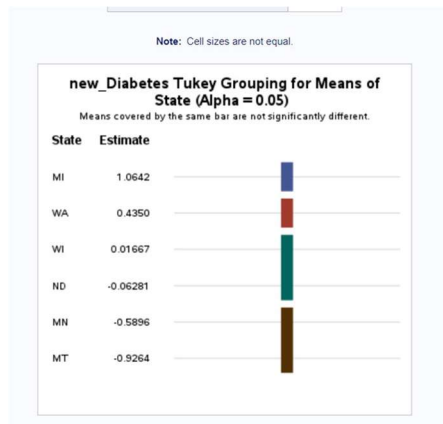
Furthermore, since the p-value in our individual MANOVA tables for both Diabetes and Cancer is less than **0.0001**, we also reject the null hypothesis for each variable individually. This indicates that the Diabetes data for at least one state and the Cancer data for at least one state are different from the others.

However, this doesn't tell us which northern states are different from each other. It simply tells us that not all the state means are equal.

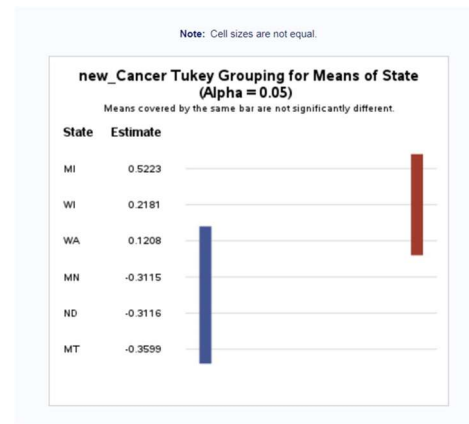
4.2 Tukey post hoc testing

(c) SAS results:

Diabetes



Cancer



Diabetes

Comparisons significant at the 0.05 level are indicated by ***.

State Comparison	Difference Between Means	Simultaneous 95% Confidence Limits		
MI - WA	0.6292	0.2217	1.0367	***
MI - WI	1.0476	0.7095	1.3857	***
MI - ND	1.1271	0.7579	1.4962	***
MI - MN	1.6538	1.3317	1.9759	***
MI - MT	1.9906	1.6276	2.3536	***
WA - MI	-0.6292	-1.0367	-0.2217	***
WA - WI	0.4184	0.0010	0.8357	***
WA - ND	0.4979	0.0550	0.9407	***
WA - MN	1.0246	0.6201	1.4292	***
WA - MT	1.3614	0.9236	1.7992	***
WI - MI	-1.0476	-1.3857	-0.7095	***
WI - WA	-0.4184	-0.8357	-0.0010	***
WI - ND	0.0795	-0.3005	0.4594	
WI - MN	0.6063	0.2718	0.9407	***
WI - MT	0.9430	0.5690	1.3171	***
ND - MI	-1.1271	-1.4962	-0.7579	***
ND - WA	-0.4979	-0.9407	-0.0550	***
ND - WI	-0.0795	-0.4594	0.3005	
ND - MN	0.5268	0.1610	0.8926	***
ND - MT	0.8636	0.4613	1.2658	***
MN - MI	-1.6538	-1.9759	-1.3317	***
MN - WA	-1.0246	-1.4292	-0.6201	***
MN - WI	-0.6063	-0.9407	-0.2718	***
MN - ND	-0.5268	-0.8926	-0.1610	***
MN - MT	0.3368	-0.0229	0.6964	
MT - MI	-1.9906	-2.3536	-1.6276	***
MT - WA	-1.3614	-1.7992	-0.9236	***
MT - WI	-0.9430	-1.3171	-0.5690	***
MT - ND	-0.8636	-1.2658	-0.4613	***
MT - MN	-0.3368	-0.6964	0.0229	

Cancer

Comparisons significant at the 0.05 level are indicated by ***.

State Comparison	Difference Between Means	Simultaneous 95% Confidence Limits		
MI - WI	0.3042	-0.1290	0.7375	
MI - WA	0.4015	-0.1207	0.9238	
MI - MN	0.8338	0.4210	1.2465	***
MI - ND	0.8340	0.3610	1.3070	***
MI - MT	0.8822	0.4170	1.3474	***
WI - MI	-0.3042	-0.7375	0.1290	
WI - WA	0.0973	-0.4375	0.6322	
WI - MN	0.5296	0.1010	0.9581	***
WI - ND	0.5297	0.0429	1.0166	***
WI - MT	0.5780	0.0987	1.0573	***
WA - MI	-0.4015	-0.9238	0.1207	
WA - WI	-0.0973	-0.6322	0.4375	
WA - MN	0.4322	-0.0862	0.9506	
WA - ND	0.4324	-0.1351	1.0000	
WA - MT	0.4806	-0.0804	1.0417	
MN - MI	-0.8338	-1.2465	-0.4210	***
MN - WI	-0.5296	-0.9581	-0.1010	***
MN - WA	-0.4322	-0.9506	0.0862	
MN - ND	0.0002	-0.4686	0.4689	
MN - MT	0.0484	-0.4125	0.5093	
ND - MI	-0.8340	-1.3070	-0.3610	***
ND - WI	-0.5297	-1.0166	-0.0429	***
ND - WA	-0.4324	-1.0000	0.1351	
ND - MN	-0.0002	-0.4689	0.4686	
ND - MT	0.0482	-0.4673	0.5638	
MT - MI	-0.8822	-1.3474	-0.4170	***
MT - WI	-0.5780	-1.0573	-0.0987	***
MT - WA	-0.4806	-1.0417	0.0804	
MT - MN	-0.0484	-0.5093	0.4125	
MT - ND	-0.0482	-0.5638	0.4673	

(d) Comments:

To determine the specific differences between groups, we conducted a Tukey post hoc test, which enables the exploration of differences in mean values among multiple groups while controlling for the family-wise error rate.

The pairwise comparisons for **Diabetes** unveiled that only the pairs MN-MT and ND-WI did not show significant differences in mean values. Conversely, all other state pairs exhibited statistically significant differences.

Similarly, in the realm of **Cancer** rates, pairwise comparisons indicated significant differences in mean values for the pairs MI-MN, MI-ND, MI-MT, WI-MN, WI-MT, and WI-ND.

These findings suggest notable disparities in Diabetes and Cancer rates among northern states, likely influenced by factors such as demographic characteristics, healthcare policies, and environmental conditions. Notably, while Diabetes rates showed more pronounced variability across states, Cancer rates displayed comparatively less variability. This distinction hints at the potential influence of additional factors, such as dietary habits or genetic predispositions, which merit further exploration in future research endeavors.

5. Principal Component Analysis

(a) Problem:

Using **proc factor** in SAS and varimax rotation, perform a *principal components analysis* on the variables sixtyfiveandup, Smoking_Rate, physical_inactivity, Diabetes, Heart_Disease Cancer Glyphosates, NATA_Cancer_11, Mercury_TPY, and Lead_TPY for Washington state (WA).

(b) SAS program:

```
/* 5.1 Creating data set for PCA testing including WA state information only */
data Altz_PCA;
    set Altz_transformed;
    z = _N_;
    if z ge 280 and z le 318 then state = 'WA';
    if state NOT IN ("WA") then delete;
run;
/*print the data to the screen to see if they are okay;
proc print data=Altz_PCA;
run;
/* 5.2 PCA testing WA state data using proc factor and varimax rotation*/
proc factor data=Altz_PCA method=prin priors=one rotate=varimax n=3 score out=pca;
    var sixtyfiveandup Smoking_Rate physical_inactivity Diabetes
        Heart_Disease Cancer Glyphosates NATA_Cancer_11 Mercury_TPY Lead_TPY;
run;

/* The corr procedure is used to calculate pairwise correlations
* between the first 3 principal components and the original variables. */
proc corr data=pca;
    var factor1 factor2 factor3 sixtyfiveandup Smoking_Rate physical_inactivity Diabetes
        Heart_Disease Cancer Glyphosates NATA_Cancer_11 Mercury_TPY Lead_TPY;
run;
```

5.1 Examine the eigenvalues and rotated factor pattern.

(c) SAS results:

The FACTOR Procedure				
Initial Factor Method: Principal Components				
Prior Communality Estimates: ONE				
Eigenvalues of the Correlation Matrix: Total = 10 Average = 1				
	Eigenvalue	Difference	Proportion	Cumulative
1	3.42811997	1.28754659	0.3428	0.3428
2	2.14057338	0.23421041	0.2141	0.5569
3	1.90636297	1.06711506	0.1906	0.7475
4	0.83924792	0.24900535	0.0839	0.8314
5	0.59024256	0.18852812	0.0590	0.8905
6	0.40171445	0.13415079	0.0402	0.9306
7	0.26756365	0.05650145	0.0268	0.9574
8	0.21106220	0.06979115	0.0211	0.9785
9	0.14127105	0.06742921	0.0141	0.9926
10	0.07384184		0.0074	1.0000

Rotated Factor Pattern			
	Factor1	Factor2	Factor3
sixtyfiveandup	-0.55929	-0.24426	0.53622
Smoking_Rate	-0.26420	0.35630	0.83259
physical_inactivity	-0.23308	0.88018	0.09091
Diabetes	-0.06449	0.80754	0.07917
Heart_Disease	0.00752	0.78813	-0.01913
Cancer	0.05448	0.40258	0.68621
Glyphosates	-0.15204	0.39552	-0.80028
NATA_Cancer_11	0.87805	-0.10484	-0.15193
Mercury_TPY	0.82717	-0.05751	0.13199
Lead_TPY	0.89410	-0.19297	-0.03318

Variance Explained by Each Factor		
Factor1	Factor2	Factor3
2.7218298	2.6046576	2.1485689

(d) Comments:

The output provides the eigenvalues and eigenvectors (loadings) for each axis. The first principal component explains 34.28% of the total sample variance. Furthermore, the first three principal components, collectively, explain 74.75%. Consequently, sample variation is summarized very well by three principal components, and we would be satisfied if we could explain about 75% of the variation. We used a method for determining the number of axes when using only those axes with eigenvalues greater than 1.0, or the first three eigenvalues.

1) Principal component 1 – PCA1

PCA1 has large loadings (or correlations) with NATA_Cancer_11, Mercury_TPY, Lead_TPY which contrast to sixtyfiveandup variables. This means that variations along PCA1 are strongly associated with air pollution measurements. High loadings on air pollution-related variables suggest that PCA1 captures variations associated with air pollution levels. This implies that areas with higher PCA1 scores may experience elevated health risks due to increased exposure to air pollutants. Thus, we could use PCA1 scores to identify regions requiring targeted interventions to mitigate air pollution and associated health risks. PCA1 accounts for 34.28% of the variability in the data, indicating that it captures a third part of the overall variability.

2) Principal component 2 – PCA2

PCA2 has large loadings on the variables physical_inactivity, Diabetes, Heart_Disease, and to a lesser extent Cancer. These variables are associated with these health indicators.

Strong correlations with these variables indicate that PCA2 reflects variations related to lifestyle and chronic disease prevalence. Areas with higher PCA2 scores may benefit from public health initiatives aimed at promoting physical activity, healthy eating habits, and chronic disease management. Tailored interventions based on PCA2 scores could help reduce the burden of chronic diseases and improve overall population health.

3) Principal component 3 – PCA3

PCA3 has large loadings on the variables Cancer, sixtyfiveandup and Smoking_Rate. These variables are opposite to Glyphosates, based on their loading values.

These results suggest that PCA3 captures variations related to smoking behaviors and cancer prevalence, potentially influenced by environmental factors. Areas with higher PCA3 scores may require targeted smoking cessation programs and cancer screening initiatives.

5.2 Examine Correlation Matrix

(c) SAS results:

Pearson Correlation Coefficients, N = 39 Prob > r under H0: Rho=0													
	Factor1	Factor2	Factor3	sixtyfiveandup	Smoking_Rate	physical_inactivity	Diabetes	Heart_Disease	Cancer	Glyphosates	NATA_Cancer_11	Mercury_TPY	Lead_TPY
Factor1	1.00000	0.00000	0.00000	-0.55929	-0.26420	-0.23308	-0.06449	0.00752	0.05448	-0.15204	0.87805	0.82717	0.89410
Factor2	0.00000	1.00000	0.00000	-0.24426	0.35630	0.88018	0.80754	0.78813	0.40258	0.39552	-0.10484	-0.05751	-0.19297
Factor3	0.00000	0.00000	1.00000	0.53622	0.83259	0.09091	0.07917	-0.01913	0.68621	-0.80028	-0.15193	0.13199	-0.03318
sixtyfiveandup	-0.55929	-0.24426	0.53622	1.00000	0.47949	0.01026	-0.05345	-0.11302	0.04812	-0.35816	-0.61237	-0.18924	-0.33479
Smoking_Rate	-0.26420	0.35630	0.83259	0.47949	1.00000	0.45845	0.31224	0.27242	0.64715	-0.45221	-0.35942	-0.15770	-0.32022
physical_inactivity	-0.23308	0.88018	0.09091	0.01026	0.45845	1.00000	0.72269	0.60775	0.29734	0.29554	-0.36832	-0.16941	-0.30472
Diabetes	-0.06449	0.80754	0.07917	-0.05345	0.31224	0.72269	1.00000	0.43637	0.28850	0.18991	-0.07421	-0.12589	-0.22070
Heart_Disease	0.00752	0.78813	-0.01913	-0.11302	0.27242	0.60775	0.43637	1.00000	0.21658	0.30423	-0.12696	0.01690	-0.12169
Cancer	0.05448	0.40258	0.68621	0.04812	0.64715	0.29734	0.28850	0.21658	1.00000	-0.30670	-0.06992	0.00617	-0.15680
Glyphosates	-0.15204	0.39552	-0.80028	-0.35816	-0.45221	0.29554	0.18991	0.30423	-0.30670	1.00000	-0.09425	-0.16694	-0.14911
NATA_Cancer_11	0.87805	-0.10484	-0.15193	-0.61237	-0.35942	-0.36832	-0.07421	-0.12696	-0.06992	-0.09425	1.00000	0.54012	0.73043
Mercury_TPY	0.82717	-0.05751	0.13199	-0.18924	-0.15770	-0.16941	-0.12589	0.01690	0.00617	-0.16694	0.54012	1.00000	0.78624
Lead_TPY	0.89410	-0.19297	-0.03318	-0.33479	-0.32022	-0.30472	-0.22070	-0.12169	-0.15680	-0.14911	0.73043	0.78624	1.00000

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(d) Comments:

We should also note that the factor loadings are the correlations between the principal components and the variables. Thus, the correlation between the Lead_TPY and **PCA 1** is about 0.894. Similarly, the correlation between Mercury_TPY (0.827), NATA_Cancer_11 (0.878) and that factor is very high. The correlation with sixtyfiveandup (-0.559) is moderate.

Similarly, **PCA 2** is correlated most strongly with physical_inactivity (0.880), Diabetes (0.808), and Heart_Disease (0.788).

Also, **PCA 3** is correlated most strongly with Smoking_Rate (0.833), Cancer (0.686), and Glyphosates (-0.800). The correlation with sixtyfiveandup (0.536) is moderate.

The interpretation of the correlation matrix aligns with the findings obtained from the rotated factor pattern, providing further insight into the relationships between principal components and variables.

Understanding Principal Component Analysis (PCA) results helps stakeholders pinpoint priority areas for intervention and resource allocation. Utilizing PCA scores to tailor interventions to specific health and environmental concerns, policymakers and public health practitioners can effectively address community needs and enhance overall population health. Ongoing monitoring of PCA scores aids in evaluating intervention effectiveness and identifying emerging health trends, facilitating evidence-based decision-making and proactive public health strategies.

6. Factor Analysis

(a) Problem:

Using **proc factor** and varimax rotation, perform a *factor analysis using the principal factor* method on the desirable variables.

(b) SAS program:

```
/* 6. FA testing WA state data using proc factor and varimax rotation*/
proc factor data=Altz_PCA method=principal priors=smc rotate=varimax n=3 score out=fa;
  var sixtyfiveandup Smoking_Rate physical_inactivity Diabetes
    Heart_Disease Cancer Glyphosates NATA_Cancer_11 Mercury_TPY Lead_TPY;
run;

/* The corr procedure is used to calculate pairwise correlations
* between the first 3 principal components and the original variables. */
proc corr data=fa;
  var factor1 factor2 factor3 sixtyfiveandup Smoking_Rate physical_inactivity Diabetes
    Heart_Disease Cancer Glyphosates NATA_Cancer_11 Mercury_TPY Lead_TPY;
run;
```

6.1 Examine the eigenvalues and rotated factor pattern.

(c) SAS results:

Eigenvalues of the Reduced Correlation Matrix: Total = 6.80434224 Average = 0.68043422				
	Eigenvalue	Difference	Proportion	Cumulative
1	3.16154011	1.35833459	0.4646	0.4646
2	1.80320552	0.22502005	0.2650	0.7296
3	1.57818547	1.06857019	0.2319	0.9616
4	0.50961528	0.31763815	0.0749	1.0365
5	0.19197713	0.19573858	0.0282	1.0647
6	-.00376145	0.02040463	-0.0006	1.0641
7	-.02416608	0.07851686	-0.0036	1.0606
8	-.10268295	0.03513246	-0.0151	1.0455
9	-.13781541	0.03393996	-0.0203	1.0252
10	-.17175538		-0.0252	1.0000

3 factors will be retained by the NFACTOR criterion.

Factor Pattern			
	Factor1	Factor2	Factor3
sixtyfiveandup	0.43910	-0.60002	0.18083
Smoking_Rate	0.69380	-0.09755	0.60674
physical_inactivity	0.70501	0.57679	0.01447
Diabetes	0.51166	0.55718	0.07071
Heart_Disease	0.39268	0.52974	0.02293
Cancer	0.40815	0.11936	0.54035
Glyphosates	0.03965	0.44884	-0.66007
NATA_Cancer_11	-0.73848	0.37929	0.27911
Mercury_TPY	-0.54459	0.27538	0.46497
Lead_TPY	-0.75117	0.27296	0.38711

The FACTOR Procedure
Rotation Method: Varimax

Orthogonal Transformation Matrix			
	1	2	3
1	-0.72182	0.58525	0.36940
2	0.48264	0.80823	-0.33738
3	0.49601	0.06524	0.86586

Rotated Factor Pattern			
	Factor1	Factor2	Factor3
sixtyfiveandup	-0.51685	-0.21617	0.52122
Smoking_Rate	-0.24693	0.36679	0.81455
physical_inactivity	-0.22333	0.87973	0.07837
Diabetes	-0.06534	0.75438	0.06225
Heart_Disease	-0.01639	0.65947	-0.01381
Cancer	0.03102	0.37059	0.57837
Glyphosates	-0.13939	0.34290	-0.70831
NATA_Cancer_11	0.85455	-0.10743	-0.15910
Mercury_TPY	0.75663	-0.06582	0.10852
Lead_TPY	0.86597	-0.19376	-0.03439

(d) Comments:

The choice of how many common factors should be included in the model can be based on the estimated eigenvalues in much the same manner as with principal components.

The first three components explain 96.16% of the variation. Besides, the first three eigenvalues, $\lambda^1=3.16$, $\lambda^2=1.8$ and $\lambda^3=1.58$, are the only eigenvalues greater than 1.0. Moreover, three common factors will account for a cumulative proportion. We consider this to be sufficient for the current example and will base future analyses on three components.

Factor 1 is correlated most strongly with NATA_Cancer_11, Mercury_TPY, Lead_TPY which contrast to sixtyfiveandup. We can say that the first factor is primarily measuring variables related to air pollution and cancer risk. Areas with higher Factor 1 scores may experience increased health risks associated with exposure to environmental pollutants. Identifying and addressing sources of pollution can help mitigate these health risks and improve community well-being.

Similarly, **Factor 2** is correlated most strongly with physical_inactivity, Diabetes, Heart_Disease. We can say that the second factor is primarily a measure of lifestyle-related factors and chronic disease prevalence. Communities with higher Factor 2 scores may benefit from interventions promoting physical activity, healthy eating habits, and chronic disease management.

And **Factor 3** is correlated most strongly with Smoking_Rate, Cancer, sixtyfiveandup, and Glyphosates. When, Glyphosates contrast to Smoking_Rate, Cancer, sixtyfiveandup. We can say that Factor 3 captures variations in smoking behaviors, cancer prevalence, and environmental exposures. Areas with higher Factor 3 scores may require comprehensive tobacco control measures, cancer prevention strategies, and environmental regulations to reduce exposures to harmful substances.

The interpretation above is very similar to that obtained in the standardized principal component analysis.

6.2 Examine Correlation Matrix

(c) SAS results:

Pearson Correlation Coefficients, N = 39 Prob > r under H0: Rho=0													
	Factor1	Factor2	Factor3	sixtyfiveandup	Smoking_Rate	physical_inactivity	Diabetes	Heart_Disease	Cancer	Glyphosates	NATA_Cancer_11	Mercury_TPY	Lead_TPY
Factor1	1.00000	-0.02801 0.8656	-0.02944 0.8588	-0.54396 0.0003	-0.25988 0.1101	-0.23504 0.1498	-0.06876 0.6775	-0.01725 0.9170	0.03264 0.8436	-0.14671 0.3728	0.89937 <.0001	0.79632 <.0001	0.91139 <.0001
Factor2	-0.02801 0.8656	1.00000	0.01619 0.9221	-0.22956 0.1598	0.38950 0.0142	0.93421 <.0001	0.80110 <.0001	0.70031 0.0132	0.39354 0.0132	0.36414 0.0227	-0.11409 0.4892	-0.06990 0.6724	-0.20576 0.2089
Factor3	-0.02944 0.8588	0.01619 0.9221	1.00000	0.56042 0.0002	0.87581 <.0001	0.08426 0.6101	0.06693 0.6856	-0.01485 0.9285	0.62186 <.0001	-0.76158 <.0001	-0.17106 0.2978	0.11668 0.4793	-0.03698 0.8231
sixtyfiveandup	-0.54396 0.0003	-0.22956 0.1598	0.56042 0.0002	1.00000	0.47949 0.0020	0.01026 0.9506	-0.05345 0.7466	-0.11302 0.4933	0.04812 0.7711	-0.35816 0.0252	-0.61237 <.0001	-0.18924 0.2486	-0.33479 0.0372
Smoking_Rate	-0.25988 0.1101	0.38950 0.0142	0.87581 <.0001	0.47949 0.0020	1.00000	0.45845 0.0033	0.31224 0.0530	0.27242 0.0934	0.64715 <.0001	-0.45221 0.0038	-0.35942 0.0246	-0.15770 0.3377	-0.32022 0.0469
physical_inactivity	-0.23504 0.1498	0.93421 <.0001	0.08426 0.6101	0.01026 0.9506	0.45845 0.0033	1.00000	0.72269 <.0001	0.60775 <.0001	0.29734 0.0660	0.29554 0.0678	-0.36832 0.0211	-0.16941 0.3025	-0.30472 0.0593
Diabetes	-0.06876 0.6775	0.80110 <.0001	0.06693 0.6856	-0.05345 0.7466	0.31224 0.0530	0.72269 <.0001	1.00000	0.43637 0.0055	0.28850 0.0749	0.18991 0.2469	-0.07421 0.6535	-0.12589 0.4451	-0.22070 0.1770
Heart_Disease	-0.01725 0.9170	0.70031 <.0001	-0.01485 0.9285	-0.11302 0.4933	0.27242 0.0934	0.60775 <.0001	0.43637 0.0055	1.00000	0.21658 0.1854	0.30423 0.0597	-0.12696 0.4412	0.01690 0.9187	-0.12169 0.4605
Cancer	0.03264 0.8436	0.39354 0.0132	0.62186 <.0001	0.04812 0.7711	0.64715 <.0001	0.29734 0.0660	0.28850 0.0749	0.21658 0.1854	1.00000	-0.30670 0.0576	-0.06992 0.6723	0.00617 0.9703	-0.15680 0.3405
Glyphosates	-0.14671 0.3728	0.36414 0.0227	-0.76158 <.0001	-0.35816 0.0252	-0.45221 0.0038	0.29554 0.0678	0.18991 0.2469	0.30423 0.0597	-0.30670 0.0576	1.00000	-0.09425 0.5682	-0.16694 0.3097	-0.14911 0.3650
NATA_Cancer_11	0.89937 <.0001	-0.11409 0.4892	-0.17106 0.2978	-0.61237 <.0001	-0.35942 0.0246	-0.36832 0.0211	-0.07421 0.6535	-0.12696 0.4412	-0.06992 0.6723	-0.09425 0.5682	1.00000	0.54012 0.0004	0.73043 <.0001
Mercury_TPY	0.79632 <.0001	-0.06990 0.6724	0.11668 0.4793	-0.18924 0.2486	-0.15770 0.3377	-0.16941 0.3025	-0.12589 0.4451	0.01690 0.9187	-0.00617 0.9703	-0.16694 0.3097	0.54012 0.0004	1.00000	0.78624 <.0001
Lead_TPY	0.91139 <.0001	-0.20576 0.2089	-0.03698 0.8231	-0.33479 0.0372	-0.32022 0.0469	-0.30472 0.0593	-0.22070 0.1770	-0.12169 0.4605	-0.15680 0.3405	-0.14911 0.3650	0.73043 <.0001	0.78624 <.0001	1.00000

(d) Comments:

Examination of the correlation matrix reveals strong correlations between factors and variables. For instance, Factor 1 shows high correlations with Lead_TPY (0.911), Mercury_TPY (0.796), and NATA_Cancer_11 (0.899), with a moderate correlation observed with sixtyfiveandup (-0.544).

Similarly, **Factor 2** is correlated most strongly with physical_inactivity (0.934), Diabetes (0.801), and Heart_Disease (0.700).

Factor 3 demonstrates strong correlations with Smoking_Rate (0.876), Cancer (0.622), and Glyphosates (-0.762), with a moderate correlation observed with sixtyfiveandup (0.560).

Interpretation of Factor Analysis (FA) results provides valuable insights into the underlying factors influencing health and environmental outcomes. By understanding these relationships, public health practitioners can develop targeted interventions and policies to address community health needs effectively and promote overall well-being.

Appendix: SAS codes:

```
/* Consider the data set on Alzheimer's Disease in the 6 States (MI, MN, MT, ND, WA, WI) */
```

```
* 1.1 Reading data;
```

```
Proc import datafile = '/home/u63781314/Project/Alz_mortality_data_complete (1).csv'
```

```
out=Altz dbms=csv replace;
```

```
getnames= yes;
```

```
run;
```

```
/* Sort data by states */
```

```
data Altz;
```

```
set Altz;
```

```
State=scan(County, 2, ',');
```

```
County=scan(County, 1, ',');
```

```
z = _N_;
```

```
if z ge 1198 and z le 1280 then state = 'MI';
```

```
if z ge 1281 and z le 1367 then state = 'MN';
```

```
if z ge 1565 and z le 1620 then state = 'MT';
```

```
if z ge 1957 and z le 2009 then state = 'ND';
```

```
if z ge 2921 and z le 2959 then state = 'WA';
```

```
if z ge 3015 and z le 3086 then state = 'WI';
```

```
if state NOT IN ("MI","MN","MT","ND","WA","WI") then delete;
```

```
run;
```

```
/* Create a new dataset with only the desired variables */
```

```
data Altz;
```

```
set Altz (keep = var1 state sixtyfiveandup Smoking_Rate physical_inactivity Diabetes
```

```
Heart_Disease Cancer Glyphosates NATA_Cancer_11 Mercury_TPY Lead_TPY);
```

```
run; /* keep only the desired variables */
```

```
*print the data to the screen to see if they are okay;
```

```
proc print data= Altz;
```

```
run;
```

```
/* 2 Univariate normality assessment */
```

```
/* 2.1 Perform univariate tests for normality */
```

```
proc univariate normal plot;
```

```
var Diabetes Cancer; run;
```

```
/* 2.2.1 Perform univariate tests for normality by states for Diabetes*/
```

```
proc sort; by state;
```

```
proc univariate normal plot;
```

```
var Diabetes; by state; run;
```

```
/* 2.2.2 Perform univariate tests for normality by states for Cancer*/
```

```
proc sort; by state;
```

```
proc univariate normal plot;
```

```
var Cancer; by state; run;
```

```

/* 2.3 Outliers assessment */
/* Sort the data by the smallest Diabetes variable */
proc sort data=Altz;
    by Diabetes;
run;
/* Print the top and bottom few observations to identify counties with extreme small values */
proc print data=Altz(obs=10) ; /* Print the first 10 observations */
    title 'Counties with Smallest Diabetes Values';
run;
/* Sort the data by the largest Diabetes variable in descending order */
proc sort data=Altz;
    by descending Diabetes;
run;
/* Print the top and bottom few observations to identify counties with extreme large values */
proc print data=Altz(obs=10); /* Print the first 10 observations */
    title 'Counties with Largest Diabetes Values';
run;

/* 2.4 Bivariate tests for normality */

/* Generate scatterplots for pairs of variables */;
proc corr data=Altz plots(only)=scatter;
    var Diabetes Cancer;
run;
* Chi-Square and Gamma plots for Diabetes and Cancer (bivariate normality);
proc princomp std out=pcresult;var Diabetes Cancer;
run;
data mahal;set pcresult;dist2=uss(of prin1-prin2);
run;
proc sort;by dist2;run;
data plotdata;set mahal;
prb=(_n_ -.5)/390;chiquant=cinv(prb,2);
run;
proc gplot;plot dist2*chiquant;
run;

/* 3 Finding transformation for normality */

/* 3.1.1 perform box-cox transformation for Diabetes*/
proc rank normal=blom out=normals;
    var Diabetes; ranks Dq;
proc transreg;
    model boxcox(Diabetes) = identity(Dq);
run;
/* Apply new transformation*/
data Altz;
set Altz;
n_Dia = (Diabetes**(-0.75)-1/(-0.75));
/*print the data to the screen to see if they are okay */
proc print data= Altz;
run;

```

```

/* Check for normality after transformation */
proc univariate normal plot;
var n_Dia; run;

/*3.1.2 perform box-cox transformation for Cancer*/
proc rank normal=blom out=normals;
var Cancer; ranks Cq;
proc transreg;
model boxcox(Cancer) = identity(Cq);
run;

/* 3.2 Quantile transformation for variables */
proc rank data=Altz normal=blom out=normals;
var Diabetes;
ranks new_Diabetes;
var Cancer;
ranks new_Cancer;
run;
/* Print transformed data */
proc print data=normals;
run;
/* Perform univariate tests for normality for transformed Diabetes */
proc univariate data=normals normal plot;
var new_Diabetes;
run;
/* Perform univariate tests for normality for transformed Cancer */
proc univariate data=normals normal plot;
var new_Cancer;
run;
/* Create new dataset with transformed variable */
data Altz_transformed;
merge Altz(in=a) normals(in=b);
run;
/* Print transformed data */
proc print data=Altz_transformed;
title 'Transformed Data';
run;

/* 4. MANOVA testing */
/* MANOVA testing for new_Diabetes and new_Cancer */
proc glm;
class state;
model new_Diabetes new_Cancer = state/ss1 ss2;
means state / tukey lines;
means state / tukey;
MANOVA h=state/printe printh ;
Run;

/* 5. Perform PCA for northern states */
* 5.1 Creating data set for PCA testing including WA state information only;

```

```

data Altz_PCA;
    set Altz_transformed;

z = _N_;
    if z ge 280 and z le 318 then state = 'WA';
if state NOT IN ("WA") then delete;
run;
*print the data to the screen to see if they are okay;
proc print data=Altz_PCA;
run;

/* 5.2 PCA testing WA state data using proc factor and varimax rotation*/
proc factor data=Altz_PCA method=prin priors=one rotate=varimax n=3 score out=pca;
    var sixtyfiveandup Smoking_Rate physical_inactivity Diabetes
        Heart_Disease Cancer Glyphosates NATA_Cancer_11 Mercury_TPY Lead_TPY;
run;
/* The corr procedure is used to calculate pairwise correlations
* between the first 3 principal components and the original variables. */
proc corr data=pca;
    var factor1 factor2 factor3 sixtyfiveandup Smoking_Rate physical_inactivity Diabetes
        Heart_Disease Cancer Glyphosates NATA_Cancer_11 Mercury_TPY Lead_TPY;
run;

/* 6. FA testing WA state data using proc factor and varimax rotation*/
proc factor data=Altz_PCA method=principal priors=smc rotate=varimax n=3 score out=fa;
    var sixtyfiveandup Smoking_Rate physical_inactivity Diabetes
        Heart_Disease Cancer Glyphosates NATA_Cancer_11 Mercury_TPY Lead_TPY;
run;
/* The corr procedure is used to calculate pairwise correlations
* between the first 3 principal components and the original variables. */
proc corr data=fa;
    var factor1 factor2 factor3 sixtyfiveandup Smoking_Rate physical_inactivity Diabetes
        Heart_Disease Cancer Glyphosates NATA_Cancer_11 Mercury_TPY Lead_TPY;
run;

```